ORF 524: Statistical Theory and Methods – Fall 2026

Princeton University

Last update: October 30, 2026

1 Introduction

This graduate class provides a theory based introduction to classical and modern mathematical statistics at the Ph.D. level. Its main focus is on foundational aspects with applications to estimation and inference methods.

2 Useful Information

Instructor:	Matias D. Cattaneo (cattaneo@princeton.edu)
Lectures:	Mon and Wed, 2:55 pm – 4:15 pm, Sherrerd 101
Class Website:	https://canvas.princeton.edu/
GitHub:	https://github.com/mdcattaneo/ORF-524
TA:	Thierry Rioual (tr4334@princeton.edu)
Precepts:	Tue, 5:30 pm – 6:50 pm, Sherrerd 001

3 Grading

Assessment will be based on two in class, closed book midterm exams, each worth 50% of the final numerical grade. The final letter grade will be determined based on overall performance in the class. A **tentative** mapping from the final numerical grade to the final letter grade is available by default in Canvas, following the recommended grading practices and policies at Princeton University. I reserve the right to alter the mapping between the final numerical grade and final letter grade without notice.

4 Course Format and Use of AI

ORF 524 is an instructor led, AI integrated course. Each teaching week is organized around a Markdown chapter available in the course GitHub repository. These chapters are the stable mathematical spine for preparation, class, and review; they do not replace lecture attendance and discussion. During class, I will use each file to navigate the material, develop important arguments on my iPad or blackboard, and at selected points interact with an AI system in view of the class. We will use generated explanations, proof attempts, examples, and counterexamples as objects to inspect rather than as automatically correct answers. I remain responsible for the course narrative, pace, and mathematical judgment.

Students are encouraged to use AI while preparing for class, studying the Markdown chapters,

and working on the ungraded practice assignments also available in the course GitHub repository. Productive uses include retrieving prerequisites, requesting a staged hint, comparing solution strategies, finding the first gap in an argument, generating a nearby practice question, and reconstructing a solution after an attempt. Students should make a genuine attempt before requesting a complete solution and should preserve regular opportunities to work without assistance. The repository's course AI guide and folder-level guides provide reusable prompts and a staged study protocol. These guides support learning but do not override this syllabus or an explicit instruction given in class.

AI output is neither an empirical fact nor a scholarly or mathematical authority. Students are responsible for checking its assumptions, derivations, and conclusions against the course references, their own reasoning, and other reliable cited sources. Generated material must not be represented as the student's independent reasoning or as an instructor approved solution. If work is ever submitted or shared for evaluation outside the two midterms, the instructions for that activity will specify whether AI is permitted and what disclosure is required.

The two midterm exams are closed book and unaided, except that each student may use one sheet of personal notes (both sides). AI systems, electronic devices, other notes, books, and outside assistance may not be used except for an accommodation authorized by the University. The exams assess the same standard of independent mathematical mastery expected before generative AI: students must be able to state assumptions, select and justify methods, reconstruct arguments, and defend conclusions on their own. After the exam period has ended, a device may be used solely to scan and upload solutions during the stated Gradescope upload period.

No required learning activity depends on a paid AI service or on creating an account with a third party. Every required activity can be completed without AI through the chapters, lectures, precepts, instructor or TA support, and references. Students should not enter private student work, grades, accommodations, personal information, or restricted course copyrighted material into an external AI system. Princeton University policies and specific course instructions concerning academic integrity, privacy, and intellectual property apply throughout.

5 Main References

(1) Shao, J. (2003): Mathematical Statistics, Springer.

(2) Keener, R. W. (2010): Theoretical Statistics: Topics for a Core Course, Springer.

(3) Newey, W. K. and D. McFadden (1994): *Large Sample Estimation and Hypothesis Testing*, Handbook of Econometrics IV, North Holland.

6 Other References

(1) Bickel, P. J. and K. A. Doksum (2015): Mathematical Statistics: Basic Ideas and Selected Topics (vol. I, 2nd ed.), CRC.

(2) Bickel, P. J. and K. A. Doksum (2016): Mathematical Statistics: Basic Ideas and Selected Topics (vol. II), CRC.

(3) Eggermont, P. P. B. and V. N. LaRiccia (2001): Maximum Penalized Likelihood Estimation, Volume I: Density Estimation, Springer.

(4) Eggermont, P. P. B. and V. N. LaRiccia (2009): Maximum Penalized Likelihood Estimation, Volume II: Regression, Springer.

(5) Györfi, L. et al. (2002): A Distribution-Free Theory of Nonparametric Regression, Springer.
(6) Lehmann, E. L. and G. Casella (1998): Theory of Point Estimation (2nd ed.), Springer.
(7) Lehmann, E. L. and J. P. Romano (2005): Testing Statistical Hypotheses (3rd ed.), Springer.
(8) Severini, T. A. (2005): Elements of Distribution Theory, Cambridge.
(9) van der Vaart, A. W. (1998): Asymptotic Statistics, Cambridge.
(10) Wainwright, M. J. (2019): High-Dimensional Statistics: A Non-Asymptotic Viewpoint, Cambridge.

7 Weekly Schedule and Important Dates

Week 1 (Wed 2-Sep):	Introduction.
Week 2 (Wed 9-Sep):	Statistical Foundations.
Week 3 (Mon 14-Sep):	Statistical Decision Theory and Point Estimation.
Week 4 (Mon 21-Sep):	Hypothesis Testing and Uncertainty Quantification.
Week 5 (Mon 28-Sep):	Convergence and Limit Theorems.
Week 6 (Mon 5-Oct):	Delta Method and Asymptotic Inference.
Week 7 (Mon 12-Oct):	**Midterm Exam 1**.
Week 8 (Mon 19-Oct):	*Fall break*.
Week 9 (Mon 26-Oct):	M and Z Estimation.
Week 10 (Mon 2-Nov):	Regression Applications and Two Step Estimation.
Week 11 (Mon 9-Nov):	Kernel Based Nonparametric Methods.
Week 12 (Mon 16-Nov):	Series Based Nonparametric Methods.
Week 13 (Mon 23-Nov):	*Thanksgiving Break*.
Week 14 (Mon 30-Nov):	Semiparametric Methods.
Week 15 (Mon 7-Dec):	**Midterm Exam 2**.

- **Midterm exam 1**: Mon 12–Oct, 2:55 pm, 80 minutes, in person, closed book, one sheet of personal notes (both sides). 10 extra minutes to upload solutions to Gradescope.
- **Midterm exam 2**: Mon 7–Dec, 2:55 pm, 160 minutes, Friend 008, in person, closed book, one sheet of personal notes (both sides). 10 extra minutes to upload solutions to Gradescope.